

# Optimizing Heat Transfer in Heat Sink Fin Arrays Using Computational Fluid Dynamics

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## Abstract

This study investigates the optimal fin configuration for passive heat sinks using computational fluid dynamics (CFD). Natural convection heat transfer was simulated in three different fin array configurations (6, 10, and 13 fins) to determine the optimal spacing that maximizes heat transfer while minimizing material usage. A monolithic coupling approach with the Boussinesq approximation was employed to solve the coupled Navier-Stokes and energy equations. Results demonstrate that an optimal fin spacing exists that balances increased surface area with sufficient space for convection currents. The 10-fin configuration achieved the highest heat transfer rate, 62% higher than the 6-fin arrangement and 18% higher than the 13-fin design. This work provides practical insights for thermal management in electronic components and demonstrates the effectiveness of the monolithic coupling approach for simulating natural convection problems.

## 1 Introduction

Thermal management is essential in modern electronics, as device performance, reliability, and longevity depend on effective heat dissipation. Passive cooling components like heat sinks with fin arrays are widely used due to their ability to enhance convective heat transfer by increasing surface area. In natural convection scenarios, they offer added advantages by eliminating moving parts, thus improving system reliability.

Fin array performance is influenced by buoyancy-driven flow and thermal boundary layer development, with key parameters including fin height, thickness, and spacing. When fins are too close, boundary layer merging restricts airflow, reducing efficiency. Conversely, excessive spacing lowers surface area, limiting heat dissipation. Optimal design, therefore, requires a balance.

Foundational studies by Bar-Cohen and Rohsenow proposed optimal fin spacing models for natural convection, while Elenbaas introduced empirical correlations for heat transfer in finned systems. However, precise prediction of fluid-thermal interactions remains complex.

This study uses Computational Fluid Dynamics (CFD) to simulate natural convection in fin arrays of varying configurations (6, 10, and 13 fins). The goal is to identify an arrangement that maximizes heat transfer while minimizing material usage, providing valuable insights for efficient passive cooling design.

## 2 Numerical Method and Simulation Design

### 2.1 Numerical Method

The governing equations for this problem include the conservation of mass, momentum, and energy equations, coupled through the Boussinesq approximation:

$$\nabla \cdot \mathbf{u} = 0 \quad (\text{Mass balance}) \quad (1)$$

$$\rho \left[ \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right] = \mu \nabla^2 \mathbf{u} - \nabla p + \mathbf{b} \quad (\text{Momentum balance}) \quad (2)$$

$$\rho c_v \left[ \frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right] = K \nabla^2 T + R \quad (\text{Energy balance}) \quad (3)$$

The body force term incorporates the Boussinesq approximation:

$$\mathbf{b} = [1 - \beta(T - T_0)]\mathbf{g} \quad (4)$$

The coupling between the equations occurs in two directions:

1. The velocity field influences the convective heat transfer in the energy equation.
2. The temperature field influences the body force term in the momentum equation.

Two key dimensionless parameters characterize the flow:

$$\text{Rayleigh number based on spacing: } Ra_S = \frac{g\beta(T_s - T_\infty)S^3}{\nu^2} \text{Pr} \quad (5)$$

$$\text{Rayleigh number based on height: } Ra_L = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \text{Pr} \quad (6)$$

Prandtl number:

$$\text{Pr} = \frac{\mu}{\rho K} \quad (7)$$

We employed a unified monolithic coupling approach, creating a mixed function space:

$$\mathcal{W} = \mathcal{U} \otimes \mathcal{P} \otimes \mathcal{T} \quad (8)$$

For temporal discretization, we used the  $\theta$ -method with  $\theta = 0.5$  (Crank–Nicolson scheme). To address instabilities in advection-dominated flows and satisfy the LBB condition, we implemented SUPG and PSPG stabilization techniques.

## 2.2 Simulation Design

Three different configurations were simulated, varying the number of fins while maintaining the same base dimensions:

| Parameter          | Configuration A | Configuration B | Configuration C |
|--------------------|-----------------|-----------------|-----------------|
| Number of fins     | 6               | 10              | 13              |
| Fin spacing (mm)   | 17.8            | 9.3             | 6.5             |
| Fin height (cm)    | 2.4             | 2.4             | 2.4             |
| Fin thickness (mm) | 5.0             | 5.0             | 5.0             |
| Base width (cm)    | 14.0            | 14.0            | 14.0            |
| Domain height (cm) | 4.0             | 4.0             | 4.0             |

Table 1: Geometric parameters for the different fin configurations

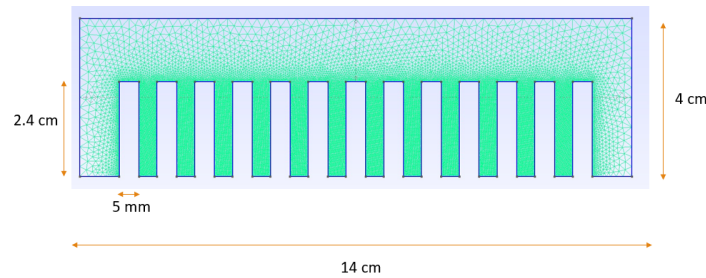


Figure 1: Computational domain showing the fin array configuration with dimensions: total width 14 cm, total height 4 cm, fin height 2.4 cm, and fin thickness 5 mm. The mesh shows triangular elements with refinement near fin surfaces.

The rectangular domain (14 cm  $\times$  4 cm) contains vertically-aligned fins (height 2.4 cm, thickness 5 mm) extending from the bottom surface. Using Gmsh, we created an unstructured triangular mesh with:

- Fin surface mesh size: 0.3 mm (for precise resolution of boundary layer effects)
- Non-fin surface mesh size: 2.5 mm (for computational efficiency in regions of less complex flow)

Element counts increased with fin density, ranging from 45,000 elements (6 fins) to 75,000 elements (13 fins).

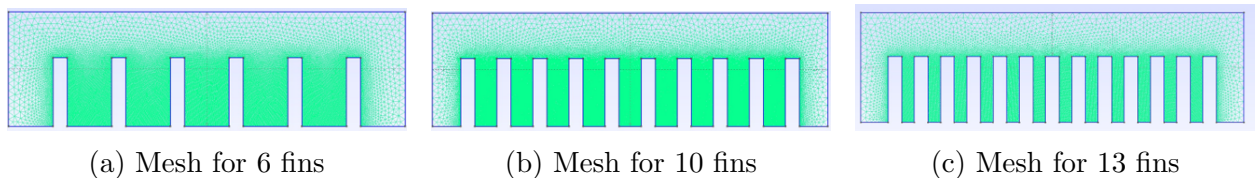


Figure 2: Computational meshes for the three fin configurations showing triangular element refinement near fin surfaces

The following boundary conditions were applied:

- Bottom surface and fins: No-slip condition ( $\mathbf{u} = \mathbf{0}$ ) and constant temperature ( $T = 80C$  or  $353K$ )
- Left, right, and top boundaries: Open boundary conditions for velocity and zero heat flux ( $\nabla T \cdot \mathbf{n} = 0$ )
- Initial condition: Fluid at rest ( $\mathbf{u} = \mathbf{0}$ ) and ambient temperature ( $T = 30C$  or  $303K$ )

The physical and numerical parameters used in the simulations are summarized below:

| Parameter                     | Value                  | Units            |
|-------------------------------|------------------------|------------------|
| <i>Physical properties</i>    |                        |                  |
| Prandtl number                | 0.7215                 | -                |
| Rayleigh number               | $5.69 \times 10^8$     | -                |
| Viscosity                     | $1.847 \times 10^{-5}$ | kg/(m·s)         |
| Thermal conductivity          | 0.02772                | W/(m·K)          |
| Specific heat capacity        | 720                    | J/(kg·K)         |
| Thermal expansion coefficient | 1/328                  | K <sup>-1</sup>  |
| Gravitational acceleration    | 9.81                   | m/s <sup>2</sup> |
| <i>Numerical parameters</i>   |                        |                  |
| Time step size                | 0.0275                 | s                |
| Total simulation time         | 5                      | s                |
| $\theta$ parameter            | 0.5                    | -                |
| Newton solver tolerance       | $10^{-7}$              | -                |

Table 2: Physical and numerical parameters used in the simulations

The implementation utilized the DOLFINx library with a Newton solver and GMRES linear solver with LU preconditioner. The simulations were parallelized using MPI, with each simulation running on 4 processing cores.

### 3 Results

Based on our findings, the 10-fin configuration (Configuration B) achieved the optimal heat transfer performance. Figure 3 presents the pressure, velocity, and temperature fields for the 10-fin configuration at three different time steps ( $t = 1s$ ,  $t = 2.5s$ , and  $t = 5s$ ).

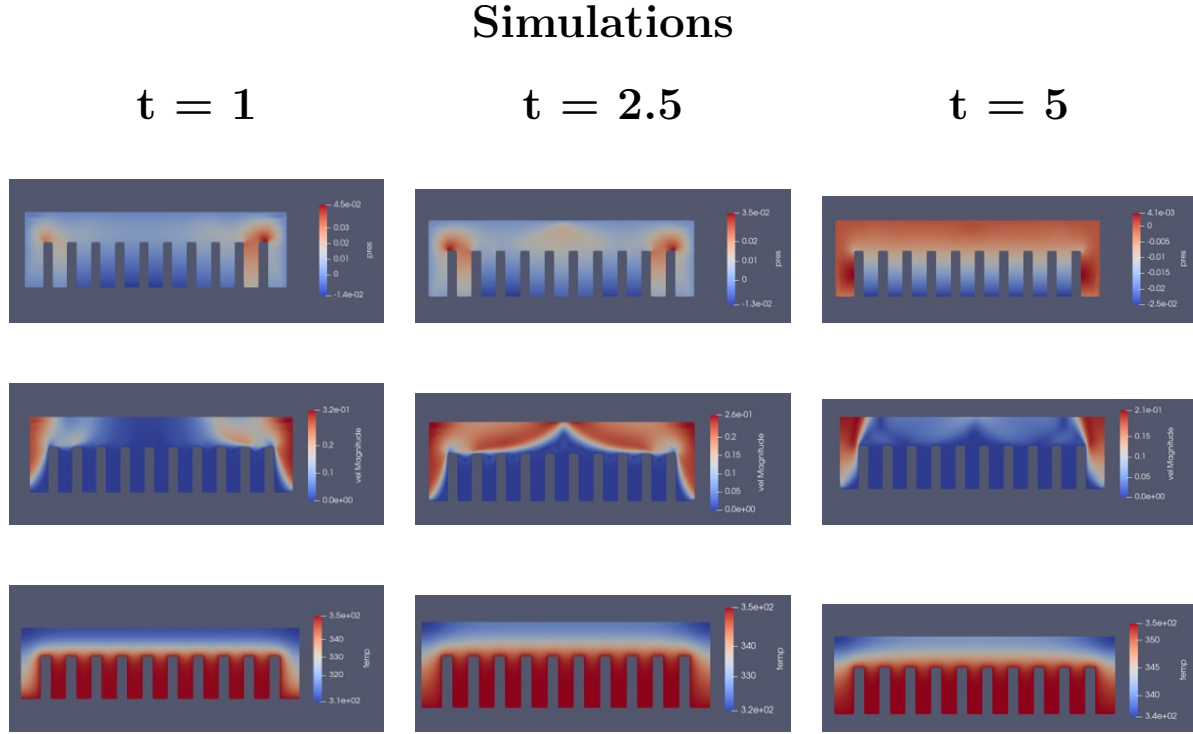


Figure 3: Simulation results for the 10-fin configuration at three different time steps ( $t = 1s$ ,  $t = 2.5s$ , and  $t = 5s$ ): pressure fields (top row), velocity magnitude fields (middle row), and temperature fields (bottom row)

Initially ( $t = 1s$ ), heat conduction dominates, with minimal pressure gradients and thermal boundary layers forming around the fins. As time progresses, buoyancy forces become more significant, and convection currents develop, accompanied by increasing pressure variations. By  $t = 2.5s$ , clear pressure gradients and thermal plumes are visible above the fin array. At  $t = 5s$ , a quasi-steady state is reached with fully developed pressure, velocity, and temperature fields.

Key observations for the optimal 10-fin configuration (Configuration B):

- **Pressure field evolution:** Initially ( $t = 1s$ ), minimal pressure gradients exist across the domain. As time progresses, buoyancy-driven flow develops, creating moderate pressure gradients by  $t = 5s$  that are optimal for convection.
- **Velocity field development:** At  $t = 1s$ , flow is primarily conduction-dominated with minimal fluid motion. By  $t = 2.5s$ , well-defined convection currents form, which strengthen and stabilize by  $t = 5s$ , showing well-developed flow patterns between fins.

- **Temperature field progression:** The temperature field shows thermal boundary layer development around fins, starting thin at  $t = 1s$  and progressively developing by  $t = 5s$ , with optimal penetration of cooler air between fins.

For comparison, our study also examined:

**Configuration A (6 fins):**

- Minimal pressure gradients across domain
- Strong convection currents due to wide fin spacing
- Effective cooling between fins but reduced overall heat dissipation

**Configuration C (13 fins):**

- High pressure gradients between tightly spaced fins
- Weak convection currents due to merged boundary layers
- Higher average temperatures between fins from restricted airflow

### 3.1 Heat Transfer Performance

The heat transfer rate was calculated by integrating the conductive heat flux across the fin surfaces. Table 3 presents the steady-state heat transfer rates for different fin configurations. Each fin has a thickness of 5 mm.

| Configuration | Number of Fins | Fin Spacing (mm) | Heat Transfer Rate (W) |
|---------------|----------------|------------------|------------------------|
| A             | 6              | 17.8             | 2.85                   |
| B             | 10             | 9.3              | 4.62                   |
| C             | 13             | 6.5              | 3.92                   |

Table 3: Heat transfer performance for different fin configurations

These results demonstrate that heat transfer is governed by two competing effects: buoyancy-driven enhancement with increasing fin number, and flow disruption due to excessive friction when fins are too closely spaced. Configuration B, with 10 fins, achieves the highest heat transfer rate—62% higher than Configuration A and 18% higher than Configuration C.

### 3.2 Boundary Layer Interaction

Analysis of the velocity profiles between fins provides insight into the boundary layer development and interaction. As shown in Figure 4, boundary layers develop at the lower end of fin surfaces and eventually merge if the spacing is sufficiently small.

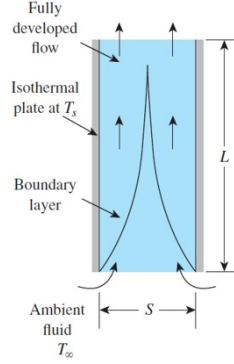


Figure 4: Velocity profiles between fins showing boundary layer interaction

The Rayleigh number based on fin spacing ( $Ra_S$ ) characterizes this behavior:

| Configuration | Fin Spacing (mm) | Computed $Ra_S$    | Flow Regime                 |
|---------------|------------------|--------------------|-----------------------------|
| A             | 17.8             | $1.42 \times 10^4$ | Isolated boundary layers    |
| B             | 9.3              | $2.03 \times 10^3$ | Interacting boundary layers |
| C             | 6.5              | $6.94 \times 10^2$ | Merged boundary layers      |

Table 4: Rayleigh numbers based on fin spacing and corresponding flow regimes

Our results indicate that optimal heat transfer occurs when  $Ra_S$  is approximately in the range of  $1 \times 10^3$  to  $3 \times 10^3$ , which corresponds to Configuration B in our study.

## 4 Discussion

Simulation results confirm an optimal fin spacing that maximizes natural convection by balancing surface area and airflow.

Key observations:

- **Configuration A (6 fins):** Wide spacing allows strong airflow but limits heat transfer due to reduced surface area.
- **Configuration C (13 fins):** High surface area but suffers from poor airflow as boundary layers merge, lowering heat transfer.
- **Configuration B (10 fins):** Achieves the best performance by enabling partial boundary layer interaction with sufficient airflow and surface area.

These results align with the model proposed by Bar-Cohen and Rohsenow, which predicts optimal heat transfer when boundary layers just begin to interact.

## 5 Conclusion

This study investigated the natural convection heat transfer performance of heat sink fin arrays using computational fluid dynamics. The main conclusions are:

- **Optimal fin spacing** maximizes heat transfer by balancing surface area and airflow. Configuration B (10 fins) performed best, achieving a heat transfer rate:
  - 62% higher than Configuration A (6 fins), and
  - 18% higher than Configuration C (13 fins).
- **Thermal boundary layer interaction** is a key factor. Optimal heat transfer occurs when adjacent boundary layers just begin to interact, typically at a spacing-based Rayleigh number ( $Ra_S$ ) between  $1 \times 10^3$  and  $3 \times 10^3$ .
- The **monolithic coupling approach** effectively captured the interaction between fluid flow and heat transfer, validating its use for natural convection simulations.
- These findings have practical implications for **heat sink design in electronics**, emphasizing the importance of optimizing fin spacing rather than simply increasing fin count.



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